

Third Terminal Examination - 2079

Class : XI

Subject: Mathematics (0071)

Full Marks: 75

Time: 3 hrs.

SET 'B'

Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group 'A'

Rewrite the best alternative of each question in your answer sheet. [11 × 1 = 11]

- If $A \subseteq B$, then $A \cap B =$
 a) A b) B c) $A - B$ d) $\emptyset$
- The set $\{x: 3 \leq x < 5\} =$
 a) $\{3, 4\}$ b) $[3, 5]$ c) $(3, 5)$ d) $[3, 5)$
- Domain of the function $f(x) = \sqrt{4 - x}$ is
 a) $(4, \infty)$ b) $(-\infty, 4)$ c) $\mathbb{R} - \{4\}$ d) $(-\infty, 4]$
- Determinant of a square matrix remains unchanged if
 a) it is inversed b) two rows are interchanged
 c) each element of a row is multiplied by a constant
 d) it is transposed
- If $z = \frac{3-4i}{\sin \theta + i \cos \theta}$, then $|z| =$
 a) $\frac{5}{\sin \theta + i \cos \theta}$ b) -1 c) 5 d) -5

13. a) If $A = \begin{bmatrix} 2 & 2 \\ 1 & -1 \end{bmatrix}$, find $A + A^T$ and then comment on the result. [2]

b) Using the properties of determinant, prove that :

$$\begin{vmatrix} b+c & a & b \\ c+a & c & a \\ a+b & b & c \end{vmatrix} = (a+b+c)(a-c)^2 \quad [3]$$

14. a) If one root of the equation $ax^2 + bx + c = 0$ be twice the other, show that $2b^2 = 9ac$. [2]

b) If the roots of the equation

$$(a^2 + b^2)x^2 - 2(ac + bd)x + (c^2 + d^2) = 0 \text{ are equal then,}$$

prove that $\frac{a}{b} = \frac{c}{d}$. [3]

15. a) Find the value of $\sin \left[\sin^{-1} \frac{4}{5} + \cot^{-1} 3 \right]$ [2]

b) Find the general solution of equation

$$\tan^2 x + (1 - \sqrt{3}) \tan x - \sqrt{3} = 0 \quad [3]$$

16. a) Find the equation of a line parallel to $5x + 12y = 0$ at a distance of 5 units from it. [2]

b) Find the equation of the bisectors of the angles between the lines

$$3x + 4y - 7 = 0 \text{ and } 12x - 5y + 14 = 0. \text{ Also, verify that bisectors are perpendicular to each other.} \quad [3]$$

17. a) Evaluate: $\lim_{x \rightarrow 2} \frac{x-2}{\ln(x-1)}$ [2]

b) Define continuity of a function at a point. Find the value of k so that

$$\text{the function } f(x) = \begin{cases} \frac{x^2-9}{x-3} & \text{if } x \neq 3 \\ 2k & \text{if } x = 3 \end{cases} \text{ is continuous at } x = 3. \quad [3]$$

18. a) Find $\frac{dy}{dx}$ when $y = \frac{1}{x - \sqrt{x^2 + a^2}}$. [2]

b) Find from first principles, the derivative of $\sin 2x$. [3]

19. a) Evaluate: $\int (e^{2x} - 1)(e^{-x} + 1)dx$ [2]

b) Evaluate: $\int \frac{x+2}{\sqrt{x-1}} dx$ [3]

Group 'C'

Long Answer Questions.

[3 × 8 = 24]

20. a) Prove that the function $f: \mathbb{R} - \{2\} \rightarrow \mathbb{R} - \{0\}$ given by

$f(x) = \frac{1}{x-2}$ is bijective. [4]

b) If z and w are two complex numbers, prove that:

$|z + w| \leq |z| + |w|$. [4]

21. a) Mention the condition that the equation

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of lines. Find the equations of two straight lines represented by the equation $2x^2 + 3xy + y^2 + 5x + 2y - 3 = 0$. Also, find the angle between the lines. [1+3+1]

b) If $A(3, 1, -3)$ and $B(4, -2, 1)$ are two points in the space, then find $\overrightarrow{AB}$, magnitude of $\overrightarrow{AB}$ and direction cosines of AB . [1+1+1]

22. a) Evaluate: $\lim_{x \rightarrow \theta} \frac{x \cos \theta - \theta \cos x}{x - \theta}$ [4]

b) Find the local maxima or local minima for the function

$f(x) = 4x^3 - 15x^2 + 12x + 7$. Also find the point of inflection. [4]

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